

# Insulin-like growth factor-1 and binding protein-3 in a 2-year soya intervention among premenopausal women.

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Soya foods may protect against the development of breast cancer. Insulin-like growth factor (IGF)-1 is under investigation as a possible link between nutrition and cancer. We examined the effect of soya foods on circulating IGF-1 and IGF binding protein (BP)-3 levels among 196 healthy premenopausal women in a 2-year randomised nutritional trial. The intervention group consumed two daily servings of soya foods including tofu, soya milk, soya nuts and soya protein powder (equivalent to 50 mg isoflavones and 5-22 g soya protein per serving); the controls maintained their regular diet. Five serum samples at baseline, 3, 6, 12, and 24 months were collected in the morning during the luteal phase and analysed for IGF-1 and IGFBP-3 by double-antibody ELISA. We applied mixed models to investigate the intervention effect and predictors of serum levels while considering the repeated measurement design. Adherence with the study regimen was high and dropout rates were acceptable.

Randomisation resulted in similar mean IGF-1 and IGFBP-3 levels by group. We did not observe a significant intervention effect on IGF-1, IGFBP-3, and their molar ratio during the entire study period. However, urinary isoflavone excretion during the study period was positively associated with IGF-1 ( $P=0.04$ ) and the IGF-1:IGFBP-3 ratio ( $P=0.06$ ). The effect was consistent over time. Adding soya foods to the diet of premenopausal women does not appear to lower serum levels of IGF-1 and IGFBP-3; if anything, the greater protein intake from soya may lead to a small increase in IGF-1 serum levels.